

Emilian Postolache, Ph.D.

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RESEARCH INTERESTS

TinyML, Speech Enhancement, Sequence Models, Deep Generative Models, Source Separation

EDUCATION

- **Sapienza University of Rome** Rome, Italy
Ph.D. in Computer Science; Advisor: Prof. Emanuele Rodolà; Final grade: Excellent with honors. 2020 - 2024
- **Sapienza University of Rome** Rome, Italy
M.Sc. in Artificial Intelligence and Robotics; Final grade: 110 with honors / 110. 2018 - 2020
- **Sapienza University of Rome** Rome, Italy
B.Sc. in Computer Science; Final grade: 110 with honors / 110. 2014 - 2018

EXPERIENCE

- **IRIS Audio Technologies** London, United Kingdom
Lead Machine Learning Researcher February 2026 - Ongoing
 - Technical lead for end-to-end ML research projects spanning speech enhancement, audio super-resolution, STT, and voice conversion with a strong focus on embedded deployment.
- **IRIS Audio Technologies** London, United Kingdom
Senior Machine Learning Researcher August 2024 - January 2026
 - Conducted research on model optimization for real-time speech enhancement targeting embedded systems.
- **Ca' Foscari University of Venice** Venice, Italy
Research Fellow March 2024 - July 2024
 - Conducted research in generative modeling applied to signal processing.
- **Sony CSL** Tokyo, Japan
AI Researcher November 2023 - February 2024
 - Worked on the task of EEG to music decoding using latent diffusion models.
- **Dolby Laboratories** Barcelona, Spain
AI Researcher June 2022 - September 2022
 - Introduced a novel method for universal sound separation, reaching state-of-the-art results. Optimized parallel experiment execution to maximize computational-resource utilization and developed modular code that enabled the team to run experiments efficiently.
- **Sapienza University of Rome** Rome, Italy
Junior Research Fellow June 2019 - May 2020
 - Conducted research in geometry processing.
- **Babelscape** Rome, Italy
Software Developer June 2018 - September 2018
 - Ported BabelNet, a very large multilingual semantic network, from Java 8 to Python 3.

RESEARCH VISITS

- **Queen Mary University of London** London, United Kingdom
Academic Visitor May 2023 - September 2023
 - Visited the Centre for Digital Music (C4DM) at Queen Mary University of London, under the supervision of Dr. Emmanouil Benetos. Worked on compositional diffusion models for music, singing voice separation, and Foley sound synthesis.

SELECTED PAPERS

- **Accelerating Transformer Inference for Translation via Parallel Decoding:** Proposed, with A. Santilli, parallel decoding methods for machine translation that provide speedups of up to 38% over greedy decoding without affecting translation quality, with mathematical guarantees, and up to 2× when scaling the available computational resources. **Accepted at ACL 2023.**
- **Multi-Source Diffusion Models for Simultaneous Music Generation and Separation:** Proposed a diffusion-based generative model capable of both waveform music synthesis and source separation. Introduced source imputation, whereby a subset of sources is generated conditioned on the others (e.g., accompaniments). Using a novel Dirac sampler, the method achieves competitive separation performance on the Slakh2100 dataset compared with state-of-the-art regressors. **Accepted at ICLR 2024 with oral presentation (top 1.2%).**
- **SyncFusion: Multimodal Onset-synchronized Video-to-Audio Foley Synthesis:** Proposed, with M. Comunità and R. F. Gramaccioni, a novel system for automating audio synthesis synchronized with video, reducing the workload of sound designers. The system uses a diffusion model conditioned on action onsets extracted from video and on text labels to generate audio. It improves both synchronization metrics and sound quality over the state-of-the-art while providing a high degree of controllability. **Accepted at ICASSP 2024.**
- **COCOLA: Coherence-Oriented Contrastive Learning of Musical Audio Representations:** Proposed COCOLA (Coherence-Oriented Contrastive Learning for Audio), the first contrastive model operating at the level of musical-track sub-constituents (stems and sub-mixtures) and capable of assessing the degree of coherence among stems. **Accepted at ICASSP 2025.**

PATENTS

- **Method for Adversarial Training for Universal Sound Separation (WO/2024/083422):** Proposed a novel *I*-replacement context-based adversarial loss and multi-discriminator training for universal sound separation (separating mixes containing any kind of sound).

PROFESSIONAL ACTIVITIES / ACADEMIC SERVICE

- **Area Chair**
 - IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), 2026
- **Session Chair**
 - IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), 2026
- **Conference Organizing Committee Member**
 - Web Chair: Smart Tools and Applications in Graphics (**STAG**), 2021
- **International Program Committee Member**
 - Interspeech Conference (**Interspeech**), 2026
 - AAAI Conference on Artificial Intelligence (**AAAI**), 2025
 - International Conference on Learning Representations (**ICLR**), 2025 - 2026
 - IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), 2024 - 2026
 - IEEE International Joint Conference on Neural Networks (**IJCNN**), 2025
 - Joint IAPR International Workshops on Statistical Techniques in Pattern Recognition and Structural and Syntactic Pattern Recognition (**S+SSPR**), 2024
 - Conference on Neural Information Processing Systems (**NeurIPS**), 2023, 2025
 - Unifying Representations in Neural Models Workshop (**UniReps**), 2023 - 2024
 - International Conference on 3D Vision (**3DV**), 2021
- **External Reviewer (not in the PC)**
 - Conference on Neural Information Processing Systems (**NeurIPS**), 2024, 2026
 - International Conference on Learning Representations (**ICLR**), 2024
 - Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2022
- **Reviewer for International Journals**
 - IEEE Signal Processing Letters (**SPL**)
 - IEEE Transactions on Visualization and Computer Graphics (**TVCG**)
 - IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**)
- **Conference Volunteering**
 - International Conference on Learning Representations (**ICLR**), 2024
- **Mentorship**
 - Micro Mentoring Experience (MiME) at IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), 2025

GRANTS AND AWARDS

- **ICASSP 2026 Practicing Engineer / Industry Professional Travel Grant (PETG/IPTG)**
IEEE Signal Processing Society *April 2026*
 - Winner of a \$1,000 travel grant awarded by the IEEE Signal Processing Society.
- **NeurIPS 2025 Top Reviewer**
NeurIPS *October 2025*
 - Granted a complimentary registration to the conference.
- **AWS Cloud Credit for Research**
Amazon *February 2024*
 - Winner, with our team, of \$50,000 worth of Amazon AWS Credit for developing research related to compositional music generation.
- **ICASSP 2024 Travel Grant**
IEEE Signal Processing Society *February 2024*
 - Winner of a \$500 travel grant awarded by the IEEE Signal Processing Society.
- **ERC Proof-of-Concept**
European Commission *May 2023*
 - Seal of excellence for a project on generative AI for music generation.
- **AAAI 2023 Student Scholarship**
Association for the Advancement of Artificial Intelligence *December 2022*
 - Winner of a \$500 scholarship awarded by the AAAI organization.
- **Imminent Grant**
Translated *April 2022*
 - Winner, with our team, of a €20,000 grant awarded by Translated for the project "Incremental Parallel Inference for Machine Translation".
- **Galileo Program**
Université Franco Italienne *January 2022*
 - Winner, with our team, of a €7,000 research grant awarded for the joint project "Multimodal Artificial Intelligence for 3D shape analysis, modelling and applications".

INVITED TALKS

- **Tech Talk: Multi-Source Diffusion Models for Simultaneous Music Generation and Separation**
PI School *21 March 2023*
 - Presented my work alongside a tutorial on diffusion models for music at the PI School of Artificial Intelligence.
- **Adversarial Permutation Invariant Training for Universal Sound Separation**
Ca' Foscari University of Venice *3 November 2022*
 - Presented my work at the Department of Environmental Sciences, Informatics and Statistics. Event organized by Prof. Luca Cosmo.

TEACHING ACTIVITIES

- Assisted Prof. M. Felisatti in the "Linear Algebra" course of the B.Sc. program "Applied Computer Science and Artificial Intelligence" (2021-2022) at Sapienza University of Rome.
- Assisted Prof. E. Rodolà in the "Metodi Numerici per l'Informatica" course of the B.Sc. program "Informatica" (2021-2022, 2022-2023) at Sapienza University of Rome.

LANGUAGES

Romanian (native), Italian (native), English (fluent), Spanish (fluent)